

A Machine Learning Model for the Prediction of Hessian Matrices based on Semiempirical Electronic Structure Theory

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Introduction

- The nuclear Hessian matrix is an important quantity in quantum chemistry used in geometry optimizations and for computing nuclear quantum effects.
- Its computation is costly and machine learning (ML) models open up opportunities to accelerate its computation significantly.
- Approach:** Develop a decision tree-based ML model using atomistic features from GFN2-xTB^[1] to predict the Hessian matrix.

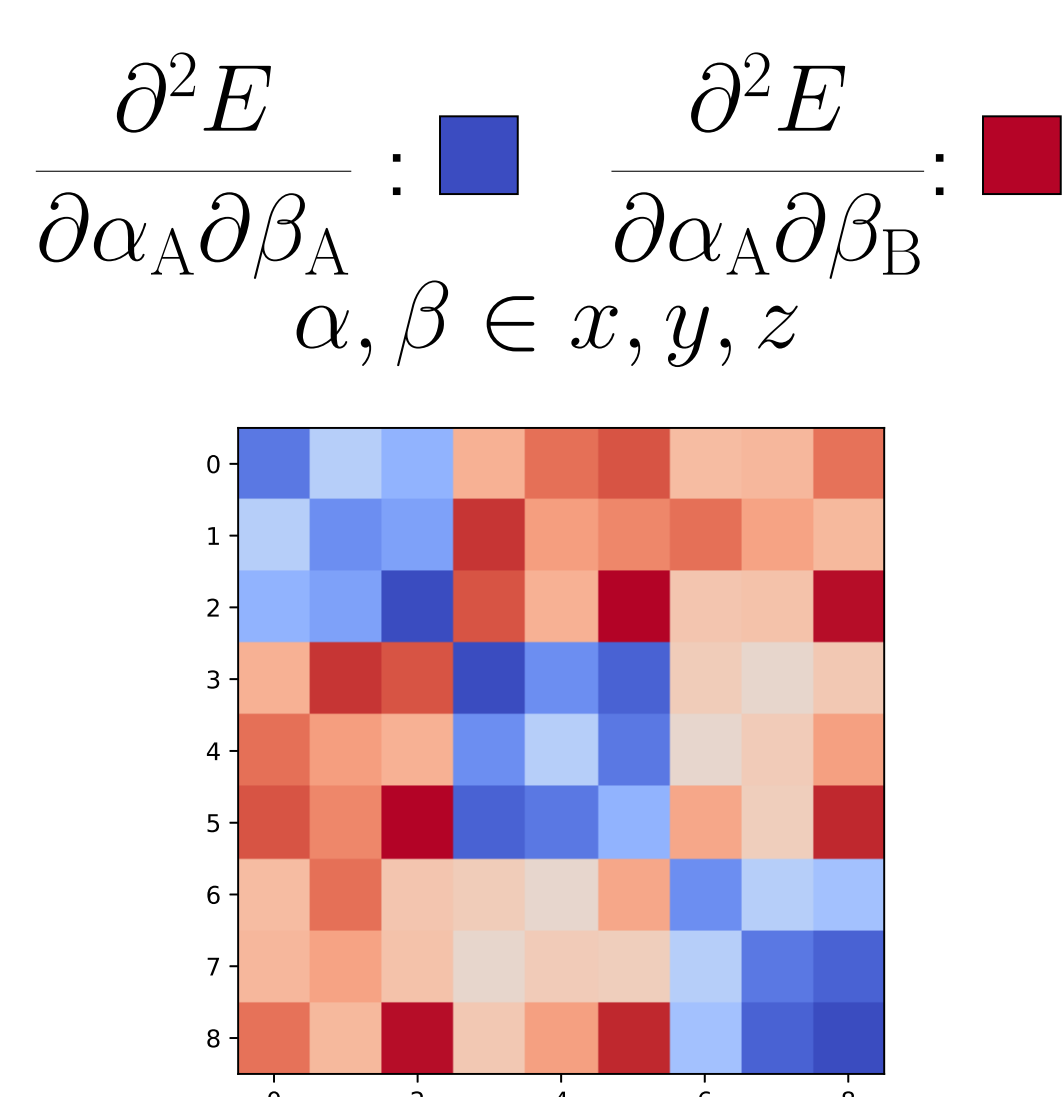
Overview of the ML Model

Employed Features

- Atomically partitioned features dependent on the geometry, electron density, and individual contributions to the total energy.
- Mulliken partitioning has been used for the generation of atomistic features.

The Hessian Matrix

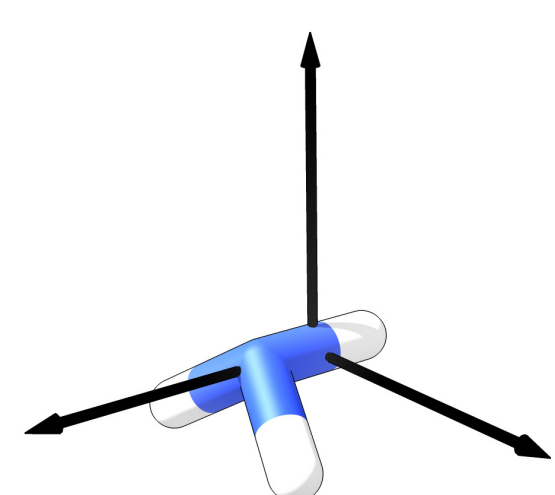
- The elements of the Hessian are second derivatives either with respect to the coordinates of a single atom or of two atoms.
- For the machine learning (ML) model, we distinguish between the diagonal blocks (blue, same atoms) and off-diagonal blocks (red, different atoms).



Connecting Features and Targets

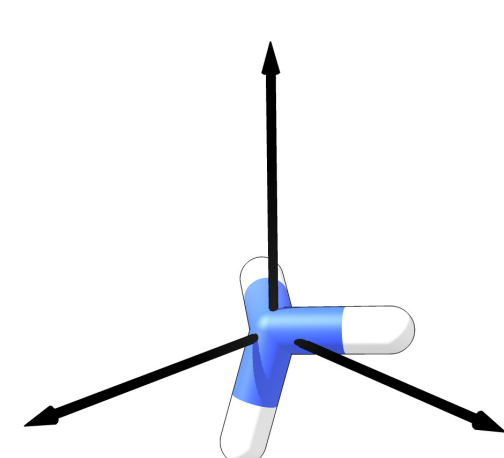
- The feature vector is either build from the quantities of one or two atoms.
- For ML modeling, a fixed feature vector length is necessary, hence two separate models are used which rely on ensemble-based decision tree methods.^[2]
- The features and targets are provided for each 3×3 block in a model-specific orientation.

Model ■ : The origin of the coordinate system is shifted to the center of two atoms and the atoms are aligned to the z-axis.



- Predicts 9 unique Hessian matrix elements.
- Uses features derived from atom A (q_A) and B (q_B) and connections of q_A and q_B ($q_A - q_B$; $q_A + q_B$; $q_A \cdot q_B$).

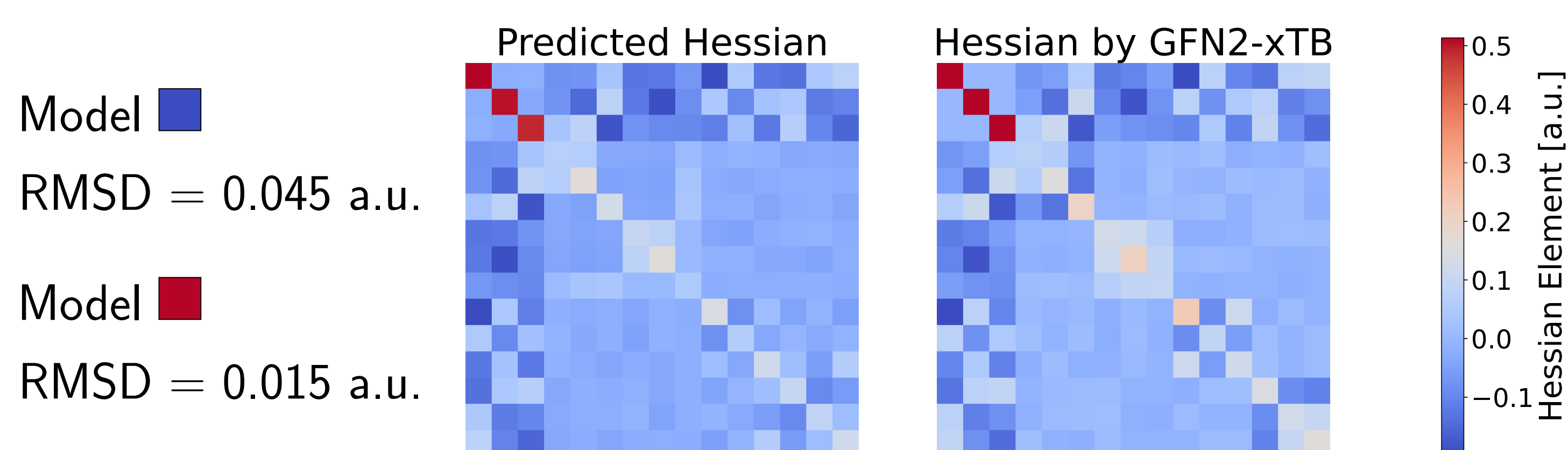
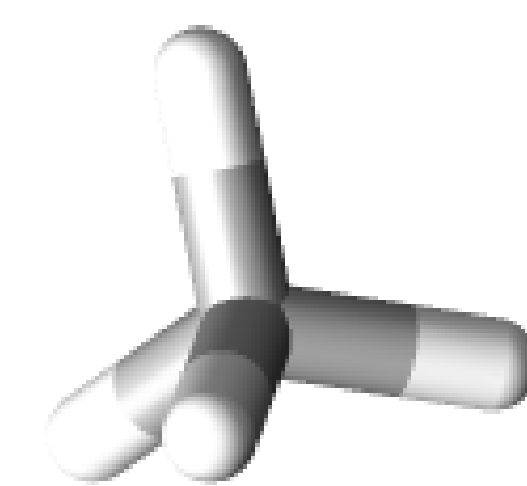
Model ■ : Here, the origin of the coordinate system is shifted to the atom.



- Predicts 6 unique Hessian matrix elements.
- Only uses features derived from atom A (q_A).

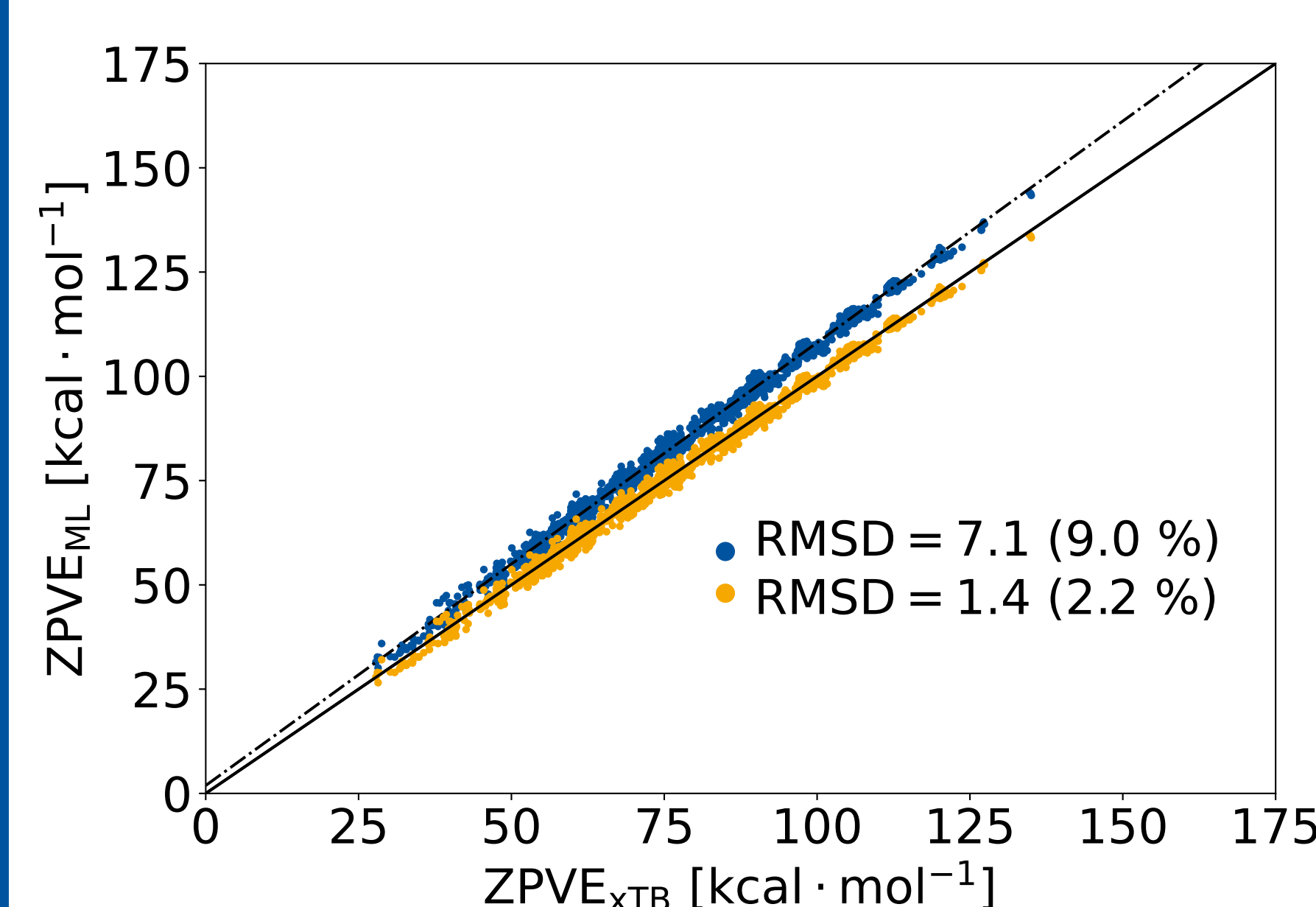
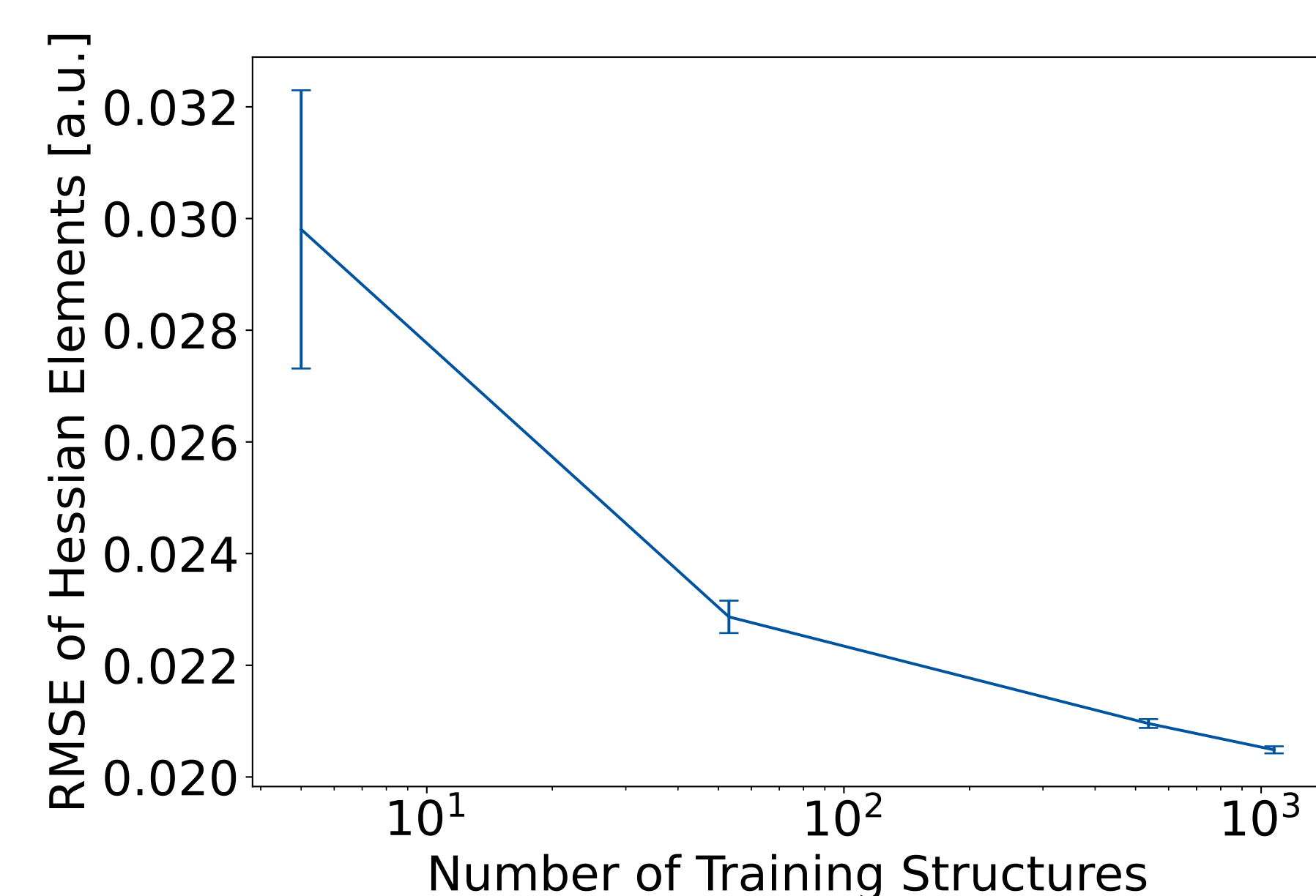
Comparison of predicted and GFN2-xTB Hessian

- Example: Hessian matrix of methane
- Overall good resemblance of the predicted Hessian compared to the computed GFN2-xTB Hessian.



Predicted ZPVE of GBD7 Set

- Learning plot computed using the GDB7^[3] set.
- The test set was fixed to 25 % of the full GBD7 set.
- The training set was increased up to 15% of the full set.
- Test and training set were randomly selected.



- Predicted zero point vibrational energy (ZPVE) for the test set with 15 % training set.
- Low scattering of the data with a small systematic overestimation of the ZPVE.
- Correcting the systematic error by a linear fit reduces RMSD significantly.

Conclusion and Future Perspective

- An ML model for the prediction of Hessian matrix elements based on atomistic features from GFN2-xTB electronic structure was developed.
- This enables the accelerated approximate computation of Hessian matrices, which can be embedded in efficient computational workflows.
- Feature enhancement for improved prediction.
- Prediction of Hessian matrices obtained at a higher level of theory, e.g. from DFT

References

- [1] C. Bannwarth, S. Ehlert, S. Grimme, *Journal of Chemical Theory and Computation* **2019**, 15, 1652.
- [2] F. Pedregosa et al., *Journal of Machine Learning Research* **2011**, 12, 2825.
- [3] M. Rupp, A. Tkatchenko, K.-R. Müller, O. A. von Lilienfeld, *Physical Review Letters* **2012**, 108.

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